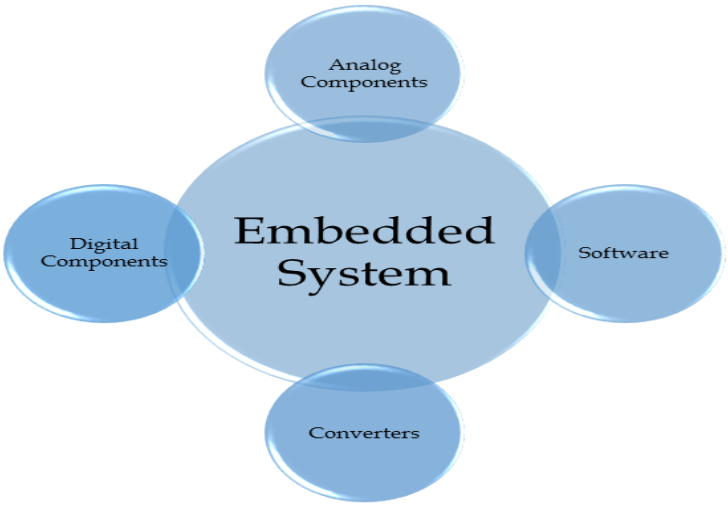
Cybersecurity in Embedded Systems

1. What are Embedded Systems?

Embedded systems are specialized computing systems designed to perform specific tasks, commonly found in IoT devices, cars, medical equipment, and smart appliances.



2. Importance of Cybersecurity

Security is critical because embedded systems often control real-world operations and can be targets for attacks.

3. Common Threats

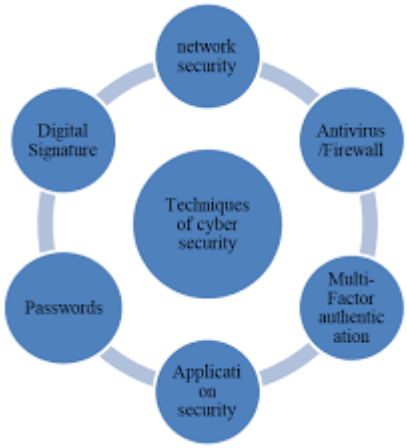
- Unauthorized access
- Malware attacks
- Data interception
- Firmware tampering
- Denial of Service (DoS)

4. Security Challenges

- Limited memory and processing power
- Lack of regular updates
- Physical access vulnerabilities

5. Security Techniques

- Encryption (AES, RSA)
- Secure Boot
- Authentication & Authorization
- Firewalls
- Intrusion Detection Systems



6. Best Practices

- Regular firmware updates
- Use secure communication protocols
- Disable unused ports
- Implement strong authentication

7. Applications

- Smart homes
- Healthcare devices
- Automotive systems
- Industrial IoT